



JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)

WEEK 1:

WEEK 1: REVISION OF SECOND TERM'S EXAMINATION

TOPIC:

Representation of Real-Life Situations on a Graph and Reasons

A. Review of Graphing Concepts

Key Points from Second Term

- Cartesian plane and coordinates
- Linear equations in two variables
- Graphing linear equations
- Reading and interpreting graphs
- Real-life applications of graphs

B. Use of Graphs for Real-Life Situations?

Advantages of Graphs

1. Visual Representation



- Easy to understand at a glance
- Clearly shows patterns and trends
- More effective than long tables

2. Quick Comparison

- Compare different data sets easily
- Shows relationships between variables
- Helps identify maximum and minimum values

3. Prediction and Estimation

- Extend lines to predict future values
- Interpolate between known points
- Helps make informed decisions

4. Communication

- Universal method of presenting data
- Useful in presentations
- Easy for a wide audience to understand

5. Problem Solving

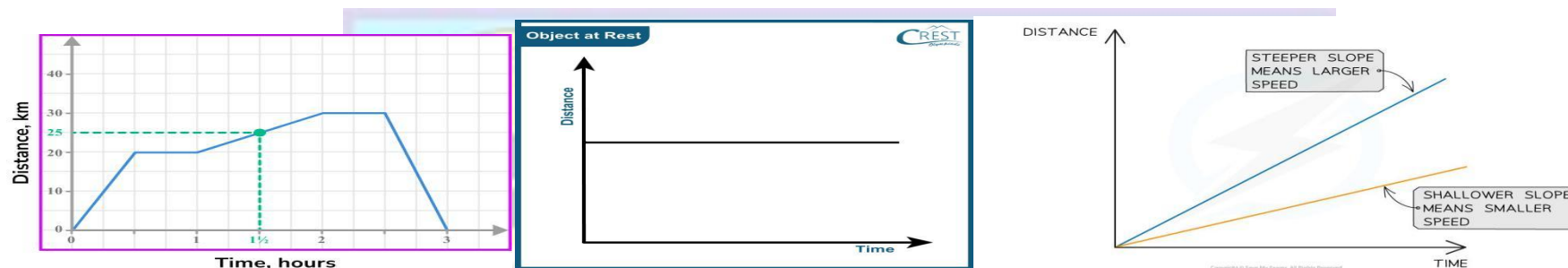
- Helps visualize solutions
 - Identifies break-even points
 - Supports better decision-making
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C. Types of Real-Life Graphs

1. Distance–Time Graphs

Shows relationship between distance and time.



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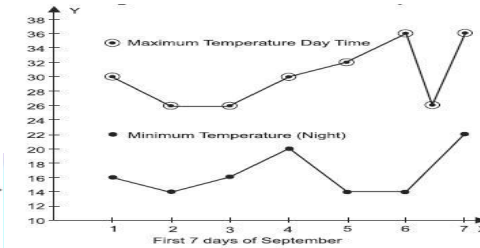
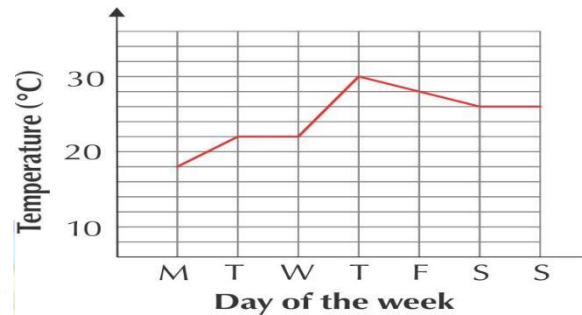
Key Interpretations:

- Slope = Speed (steeper = faster)
- Horizontal line = Stationary
- Rising line = Moving away
- Falling line = Returning

2. Temperature–Time Graphs



Shows how temperature changes over time.



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Pattern:

- Lowest at early morning
- Highest around midday
- Decreases in evening



3. Cost–Quantity Graphs

Example:

$$C = 5m + 500$$

- ₦500 = fixed cost
- ₦5 per minute = variable cost

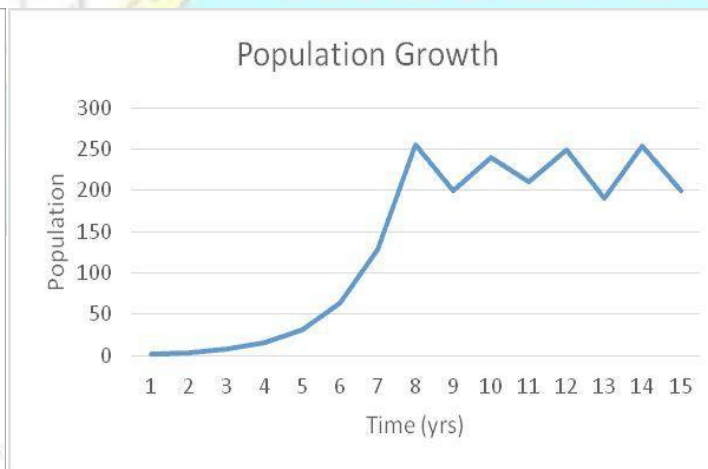
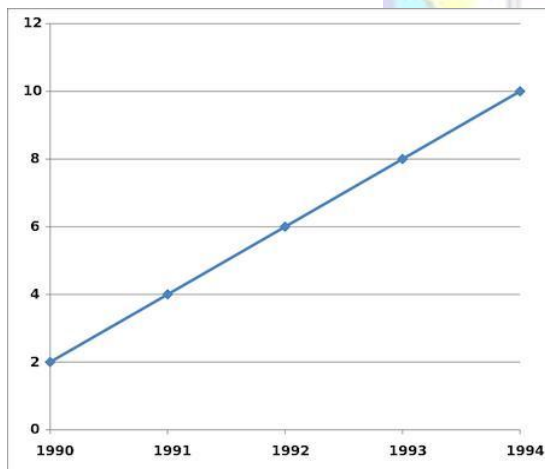
4. Profit-Loss Graphs

- Revenue: $R = 500n$
- Cost: $C = 300n + 200,000$

Break-even point:
Where Revenue = Cost

5. Growth Graphs

Shows increase over time (e.g., population growth)





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6. Conversion Graphs

Example: Currency conversion

- Straight line through origin
 - Slope = conversion rate
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D. Reading and Interpreting Graphs

Skills Required

1. Identify Variables

- X-axis → independent variable
- Y-axis → dependent variable

2. Read Values

- Find coordinates
- Interpolate (between points)
- Extrapolate (beyond data)

3. Identify Patterns





- Increasing or decreasing
- Linear or curved
- Constant or changing

4. Make Conclusions

- What does the graph show?
- What can be predicted?
- What decisions can be made?

Example: Water Tank

- Initial volume = 200 L
- Final volume = 0 L
- Time = 10 minutes
- Rate = 20 L/min

Example: Taxi Fare

$$C = 300 + 60d$$

- ₦300 = fixed charge
- ₦60 per km





E. Creating Graphs from Real-Life Data

Steps to Draw a Graph

1. Collect Data

- Identify variables
- Record values

2. Choose Scale

- Must fit data range
- Use equal intervals

3. Label Axes

- Include units

4. Plot Points

- Accurate placement

5. Draw Line/Curve

- Straight or smooth curve

6. Add Title

- Clearly describe graph





Example: Student Savings

- Monthly increase = ₦500
- Linear pattern
- Equation: $S = 500m + 1,500$

F. Practical Applications of Graphs

- **Business:** Compare profits and costs
- **Health:** Monitor patient condition
- **Education:** Track student performance
- **Finance:** Budget planning
- **Sports:** Measure performance improvement

G. Common Graph Mistakes to Avoid

- Wrong or uneven scale
 - Missing labels
 - Incorrect plotting
 - Wrong graph type
 - No title
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EVALUATION QUESTIONS

1. Give 3 advantages of graphs
2. What does a horizontal line mean?
3. How do you find speed from a graph?
4. Draw graph: ₦200 + ₦40/km
5. Find rate of temperature increase
6. What is break-even point?
7. Meaning of graph passing through origin
8. How to predict values
9. Interpolation vs extrapolation
10. Why label axes?

REVISION QUESTIONS

1. List 5 types of real-life graphs
2. Meaning of slope
3. Explain break-even point
4. How to read a graph
5. Pattern in growth graph
6. Meaning of constant term in equation
7. How graphs aid decisions
8. Purpose of conversion graphs
9. Meaning of steep slope
10. Why graphs over tables
11. Meaning of flat line
12. Choosing scale
13. Advantage of visuals



14. Can graphs predict exact future?
15. Give 3 real-life uses

ASSIGNMENT

Section A: Interpretation (25 marks)

- Distance-time graph questions:
 - Distance from home
 - Time spent
 - Speed to school
 - Speed returning
 - Total journey time

Section B: Temperature Graph (20 marks)

- Draw graph
 - Identify highest temperature
 - Check normal temperature
 - Describe pattern
 - Give conclusion
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Section C: Cost Graphs (25 marks)

1. Mobile Plan

- $\text{N}600 + \text{N}8/\text{min}$
- Draw graph
- Cost for 50 mins
- Minutes for $\text{N}1,400$

2. Compare Plans

- Plan A: $\text{N}500 + \text{N}10/\text{min}$
- Plan B: $\text{N}800 + \text{N}5/\text{min}$
- Find break-even point
- Choose cheaper option



Section D: Business Application (15 marks)

- Revenue and Cost equations
- Draw graph
- Break-even point
- Profit at 1,500 items
- Units for $\text{N}100,000$ profit